

Python Basics

Part of **Data Science Starter** · Estimated time: 45 min

Introduction

Python is the most common language for data work because its syntax stays out of the way. Before loading any real data, it helps to be comfortable with a handful of core building blocks.

Core ideas

- A list holds an ordered collection: `scores = [88, 92, 79]`. A dictionary holds key-value pairs: `student = {"name": "Ana", "score": 88}`.
- A for loop repeats an action across a collection: `for score in scores: print(score)`.
- A function packages logic for reuse: `def average(nums): return sum(nums) / len(nums)`.
- Reading a Python traceback from the bottom up shows you the actual error first (e.g. `TypeError`, `KeyError`) before the chain of calls that led to it.

Example

```
def average(nums):  
    return sum(nums) / len(nums)  
  
scores = [88, 92, 79]  
print(average(scores)) # 86.33...
```

Try it yourself

Write a function that takes a list of numbers and returns the highest and lowest values as a tuple. Test it on a list you make up, including one with negative numbers.

Further reading

- [The Python Tutorial \(official docs\)](https://docs.python.org/3/tutorial/)

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- [LearnPython.org](https://www.learnpython.org/)

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